

QUESTION	ANSWER	MARK									
1 (a)	- Graphics or images refer to digital representations of non-text information such as photographs, drawings, charts, and other visual objects - Sound is vibrations in the air that create waves of pressure such as music, spoken voice, and sound effects. - Animation is a continuous series of still images that are displayed in a sequence, one after another to simulate movement / create an illusion of motion. - Video is Product of recording moving visual images using digital camera or video devices	Select any one J2									
1 (b)	- To make multimedia application more attractive which illustrate ideas through still pictures. - To convey information more interesting, faster and easier. - To provide listening pleasure of music, special effect or the ambience of a mood setting background. - To provide interesting presentation. - To capture attention: a sound in a multimedia presentation will capture an audience's attention.	Select any one J2									
1 (c)	<table border="1"> <thead> <tr> <th></th><th>3D</th><th>4D</th></tr> </thead> <tbody> <tr> <td>Dimension</td><td>Consist of three-dimension (length, width & height)</td><td>Consist of four-dimension (length, width & height with another direction through space/motion)</td></tr> <tr> <td>Axis</td><td>X- axis , y-axis and z-axis</td><td>X- axis , y-axis and z-axis to a space</td></tr> </tbody> </table>		3D	4D	Dimension	Consist of three-dimension (length, width & height)	Consist of four-dimension (length, width & height with another direction through space/motion)	Axis	X- axis , y-axis and z-axis	X- axis , y-axis and z-axis to a space	J1,J1 J1,J1
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2 (a)	Pre-Production Elaborate ideas - What you want to do? Select team to work - Collaborate with team members Define the goal - Goals must be measurable & behaviour- based from the audience standpoint Authoring tool to use - The ‘vehicle’ for integrating all the MM elements Identify audiences - Who is the product speaking to? Delivery mode - How the product/info reach the audience - Planning – determine content, duration, incurred cost & team to work with - Resource organization – Product contents is arranged unto categories or groups - Flowchart – used to outline the content - Navigation – interacts with and moves about within a multimedia product - Define Screen Action – directs how the interface respond to the users Production Digitizing arts - Converting existing assets into digital for the use in an electronic medium	Select any one J1 Select any one J1									

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	• Modeling & animation - Models of artwork are created, rendered, and then prepared for use in the authoring tool • Shooting video - Shoot video based on the edited scripts • Recording audio - Composed audio to be used with other MM elements • Authoring - All the pieces come together in the authoring tool. The final working product is created • Proofreading - Every word on the screen is proofread and checked for consistency. In addition, video and audio are checked against the edited scripts • Quality control - Quality control goes on throughout the process. The storyboards are helpful for checking the sequencing										
2 (b)	<table border="1"> <thead> <tr> <th>Characteristics</th><th>Time-based</th><th>Icon-based</th></tr> </thead> <tbody> <tr> <td>Technique</td><td>The multimedia elements and events are <u>presented and organized along a timeline.</u></td><td>The multimedia elements and events are <u>organized as objects in a structural framework</u> or process.</td></tr> <tr> <td>Arrangement of elements</td><td>Elements are arranged in a separate layers.</td><td>Icons are dragged onto a window and linked to describe the functional flow of the application.</td></tr> </tbody> </table>	Characteristics	Time-based	Icon-based	Technique	The multimedia elements and events are <u>presented and organized along a timeline.</u>	The multimedia elements and events are <u>organized as objects in a structural framework</u> or process.	Arrangement of elements	Elements are arranged in a separate layers.	Icons are dragged onto a window and linked to describe the functional flow of the application.	J1,J1 J1,J1
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2 (c)	• Hypertext – links in text-based documents • system of managing the information related to the plain text • Hypermedia – connecting the hypertext with media such as graphics, sounds and animations • combines text-based links with graphic, audio, and video links • Tabbed Navigation – clickable area at the top that shows another page or area • Audio – control and handling multimedia applications using audio (voice search) • Hand gesture – control and handling multimedia applications through hand gesture recognition scheme • Menu – organized list of links to other web pages, navigates by pressing a key, clicking a mouse, or pressing a touch screen	Select any two J2,J2									

QUESTION	ANSWER		MARK
2 (d)	Criteria	Online	Offline
	Contents	Can be changed and updated at any time by the developer. Contents remains as original	Permanent and not changeable. Contents at a risk to damage or lost
	Accessibility	Require broadband internet connection to access multimedia contents	Can be accessed without internet connection
	capacity	Store high end multimedia contents such as video	Store limited in picture size and low resolution video
	Speed (data transfer rate)	Relatively slower data transfer rate. Subject to the constraints of the user's network.	Relatively faster data transfer rate.
	Ease of editing / updatable	Can be edited n updatable. But can be changed/damaged/ deleted by irresponsible individuals.	Can be permanently stored and are not editable. So can be quickly outdated.
3 (a)	File Processing System	Database System	Select any two J1,J1 J1,J1
	Data Redundancy – Each department has its own files in a file processing system, thus the <i>same fields are stored in multiple files.</i>	Reduced Data Redundancy – Data is only entered one time, and stored in only one file (no duplication of data).	
	Data Inconsistency – When different versions of the same data appear in different places	Data Consistency –no different versions of the same data.	
	Limited Data Sharing - It is difficult to access data stored in separate files in different departments.	Sharing of Data – Data is shared over a network, by the entire organization.	

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3 (b)	Disadvantages of DBMSs • Complexity. • DBMS is very complex piece of software, and all users must understand its functionality to take full advantage of it. • Cost of DBMS. • Can vary from only \$100 for a single-user PC, to \$100,000–\$1,000,000 for a large mainframe multi-user system. • Cost of conversion. • Cost of a DBMS may be insignificant compared with the cost of converting existing applications to run on the new DBMS and hardware (includes staff training, specialist staff to help with the conversion). • Performance. • DBMSs are general-purpose systems, and thus they may not be as efficient as some problem-specific software in some applications. • Higher impact of a failure. • Since all users and applications rely on the availability of the DBMS, the failure of any component can bring operations to a complete halt, until it is repaired.	Select any one J2
4 (a)	Data accuracy : – data values stored for an object are the correct values. Data validity : – delivery of clean and clear data to the programs, applications and services using it. Data consistency : – each user sees a reliable view of the data, including visible changes made by the user own transactions and transactions of other users.	Select any one J2
4 (b)	• Form - A window on the screen that provides areas for entering or modifying data in a database • to retrieve and maintain, view and print records in a database. • allows viewing data one record at a time. • Query - A request for specific data from the database • Specifies which records should be retrieved by specifying criteria. • Can specify the fields to be displayed. • Report - processed data that can be display on screen or printed form. • a document containing information organized in a narrative, graphic, or tabular form, prepared on ad hoc, periodic, recurring, regular, or as required basis.	Select any one J2
5(a)(i)	• Collaborative computing is a networking technology with collaborative software tools to enable individuals to work together on documents and other project components. • Videoconferencing is the two-way or multipoint reception and transmission of audio and video signals by people in different locations for real time communication.	Select any one (1) J2

QUESTION	ANSWER	MARK
5(a)(ii)	• Telemedicine can be defined as the use of network technology (computers, video, phone, messaging) by a medical professional to diagnose and treat patients. • Telesurgery a form of robot-assisted surgery (where a robot controlled by a physician operates on the patient) in which at least one of the surgeons performs the operation by controlling the robot remotely over the Internet or another network. • Monitoring system for home healthcare Electronic medical monitors take the vital signs of an individual (such as weight, bloodsugar readings, or blood pressure) or prompt an individual to answer questions (such as if he or she has eaten yet that day or has taken prescribed medication).	Select any one (1) J2
5(b)(i)	• Transmission media refers to the communication channels used to interconnect nodes on a computer network. • Transmission media refers to Materials or substances capable of carrying one or more signals in a communications channel.	Select any one (1) J2
5(b)(ii)	• Twisted Pair Cable ○ is a type of wiring in which two conductors of a single circuit are twisted together for the purposes of improving electromagnetic compatibility ○ consists of one or more twisted wires bundled together. • Coaxial cable ○ type of electrical cable consisting of an inner conductor surrounded by a concentric conducting shield, with the two separated by a dielectric. ○ Consist of a single copper wire surrounded by 3 layers • Fiber-optic cable ○ Consists of dozens or hundreds of thin strands of glass or plastic that use light to transmit signals ○ Used for high-speed communication	Select any one (1) J2
5(b)(iii)	• Cellular Radio Transmission ○ used with cell phones and are sent and received via cellular (cell) towers—tall metal towers with antennas on top. • Microwave Transmissions & Satellite Transmission ○ high-frequency radio signals that can send large quantities of data at high speeds over long distances ○ Signals can be sent or received using microwave stations or communications satellites ○ Microwave stations are earth-based stations that can transmit microwave signals directly to each other over distances of up to about 30 miles • Infrared ○ Is a wireless transmission medium that sends signals using light waves. ○ electromagnetic radiation (EMR) with wavelengths longer than those of visible light	Select any one (1) J2

QUESTION	ANSWER	MARK
5(c)(i)	Communication devices is any type of hardware capable of transmitting data, instructions and information between a sending device and a receiving device.	Select any one (1) J2
5(c)(ii)	• Network Interface Card ○ A communication device that enables a computer or device that does not have built in networking capability to access a network • Wireless Access Point ○ A central communications device that allows computers and devices to transfer data wirelessly among themselves or to a wired network using wireless technologies	J1 J1
5(c)(iii)	• DSL Modem ○ A broadband modem that sends and receives digital data over the computer to DSL line • Router ○ A communication device that connects multiple computers or other routers together on a network • Switch/hub ○ A communication device that provides a central point for cables in a network	J2 Select any one (1) J2
6(a)(i)	Search engine	J1
6(a)(ii)	Search tool X Description – software that finds web sites, web pages, images videos, news, maps or other information related to a specific topic. Search tool Y Description – Subject directory classified Web pages in an organized set of categories and related subcategory.	J2 J2
6(a)(iii)	Bing, AltaVista, Ask, Excite, Gigablast, Google, MSN, Yahoo!	J1
6(b)(i)	List search strategies - Using phrases - Using boolean operators - Using field searches	Choose any TWO J1 J1
6(b)(ii)	Using phrases -improve quality of the hits returned- essentially typing more than one keyword in a keyword search. Using boolean operators -to further specify exactly what you want to a search engine to find, using boolean (AND, OR NOT) used in keyword search. Using field searches -limits the search to a particular search characteristic (or field) such as page title, URL, page text, top level domain or web site.	J2 Choose any ONE

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6(c)(i)	Internet Messaging -Internet Messaging technology is a type of online chat allowing real-time text transmission over a network.	J2
	Forum -Forums online can be used for asking questions, making comments or initiating discussions with a large group of individuals.	J2
6(d)	Tips to improve email privacy -Treat email as a formal communication method -Don't use personal email platforms for business -Encrypt sensitive emails -Secure confidential attachments -Always enable 2-step verification	J4 Choose any TWO
6(e)	Personal application for IoT Home automation system Enable the appliances and other devices in your home such as alarm system, door locks, and sprinklers operate according to your stated preferences and can be controlled remotely via smartphone. Fitness systems Uses a sports-tested accelerometer to track your daily activity including running, walking, dancing etc in steps, time of day, and calories burned. You can set a personal goal for yourself and share your data with others via Facebook and Twitter. Smart home system Long-term benefits could include energy savings by automatically ensuring lighting, heating and air conditioning are turned off or by making the residents in the home aware of usage. Elder care system IoT can provide assistance to elderly individuals and to those with disabilities by the use of assistive technology such as voice control that can assist users with sight and mobility limitations while alert systems can be connected directly to cochlear implants worn by hearing-impaired users.	J2 Choose any ONE